

Calculus II Lesson 17

Alternating Series Test for Infinite Series $\sum_{n=1}^{\infty} (-1)^n a_n$

The alternating series $\sum_{n=1}^{\infty} (-1)^n a_n$ converges if the following conditions are satisfied:

a) $0 < a_{n+1} \leq a_n$

b) $\lim_{n \rightarrow \infty} a_n = 0$

n th Term Test for Infinite Series $\sum_{n=1}^{\infty} a_n$

a) If $\lim_{n \rightarrow \infty} a_n \neq 0$, then the series $\sum_{n=1}^{\infty} a_n$ diverges.

b) If $\lim_{n \rightarrow \infty} a_n = 0$, then the series $\sum_{n=1}^{\infty} a_n$ might converge or might diverge.

Example: $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$ and $\sum_{n=1}^{\infty} \frac{1}{n} = \infty$ (series diverges)

$\lim_{n \rightarrow \infty} \left(\frac{1}{3}\right)^n = 0$ and $\sum_{n=1}^{\infty} \left(\frac{1}{3}\right)^n = \frac{1}{2}$ (series converges)

1) $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n+4}$ is convergent or divergent?

Note: $a_n = \frac{1}{n+4}$ and $a_{n+1} = \frac{1}{n+5}$

a) Find $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{1}{n+4} = \underline{\hspace{2cm}} ?$

b) Compare $a_{n+1} \underline{\hspace{1cm}} ? \underline{\hspace{1cm}} a_n$ ($<$, $>$, or $=$)

c) What does the Alternating Series Test say about the convergent/divergent of the

alternating series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n+1}$?

2) $\sum_{n=1}^{\infty} \frac{(-1)^n}{4^n}$ is convergent or divergent?

Note: $a_n = \frac{1}{4^n}$ and $a_{n+1} = \frac{1}{4^{n+1}}$

a) Find $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{1}{4^n} =$ _____ ?

b) Compare a_{n+1} _____ ? a_n ($<$, $>$, or $=$)

c) What does the Alternating Series Test say about the convergent/divergent of the

alternating series $\sum_{n=1}^{\infty} \frac{(-1)^n}{4^n}$?

3) $\sum_{n=1}^{\infty} \frac{(-1)^n (5n-1)}{(4n+1)}$ is convergent or divergent?

a) Find $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{5n-1}{4n+1} = \underline{\hspace{2cm}} ?$

b) Use the n th Term Test to determine if the series $\sum_{n=1}^{\infty} \frac{(-1)^n (5n-1)}{(4n+1)}$ converges or diverges.

4) $\sum_{n=1}^{\infty} \frac{(-1)^n}{n!}$ is convergent or divergent?

Note: $a_n = \frac{1}{n!}$ and $a_{n+1} = \frac{1}{(n+1)!}$; therefore $a_{n+1} < a_n$

a) Find $\lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} \frac{1}{n!} = \underline{\hspace{1cm}}?$

b) Compare $a_{n+1} \underline{\hspace{1cm}}? \underline{\hspace{1cm}} a_n$ ($<$, $>$, or $=$)

c) What does the Alternating Series Test say about the convergent/divergent of the

alternating series $\sum_{n=1}^{\infty} \frac{(-1)^n}{n!}$?

5) $\sum_{n=1}^{\infty} \frac{(-1)^{n+1} \sqrt{n}}{n+4}$ is convergent or divergent?

$$\text{Note: } a_n = \frac{\sqrt{n}}{n+4} \text{ and } a_{n+1} = \frac{\sqrt{n+1}}{(n+1)+4};$$

To see that $a_{n+1} < a_n$, we can look at the graphs $y = \frac{\sqrt{x}}{x+4}$ and $y = \frac{\sqrt{x+1}}{(x+1)+4}$ for $x \geq 1$

We can see that graph of $y = \frac{\sqrt{x+1}}{(x+1)+4}$ is below the graph of $y = \frac{\sqrt{x}}{x+4}$ for $x \geq 1$, therefore $a_{n+1} < a_n$.

(Also, another way to show $a_{n+1} < a_n$ is by showing the derivative $D_x \left(\frac{\sqrt{x}}{x+4} \right) < 0$ for $x \geq 2$)

a) $D_x \left(\frac{\sqrt{x}}{x+4} \right) = \underline{\hspace{1cm}} \quad (\text{Multiple choice})$

A) $\frac{(x+4) \left(\frac{1}{2} x^{-1/2} \right) - 4\sqrt{x}}{(x+4)^2}$

B) $\frac{(x+4) \left(\frac{1}{2} x^{-1/2} \right) - 3\sqrt{x}}{(x+4)^2}$

C) $\frac{(x+4) \left(\frac{1}{2} x^{-1/2} \right) - 2\sqrt{x}}{(x+4)^2}$

D) $\frac{(x+4) \left(\frac{1}{2} x^{-1/2} \right) - \sqrt{x}}{(x+4)^2}$

b) Compare $a_{n+1} \underline{\hspace{1cm}} a_n \quad (<, >, \text{ or } =)$

c) What does the Alternating Series Test say about the convergent/divergent of the

alternating series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1} \sqrt{n}}{n+4}$

Graph $y = \frac{\sqrt{x}}{x+4}$ and $y = \frac{\sqrt{x+1}}{(x+1)+4}$ on same graph.



